



JIMMA CITY ADMINISTRATION

Functional & Non-Functional Requirements

Requirements baseline and acceptance reference

Purpose: Practical reference for implementation, operations, training, and handover of the Jimma City Digital Street Address MIS.

Document: MIS-REQ-001

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System: Jimma City Digital Street Address MIS

1. System overview

The Jimma City Digital Street Address MIS is a municipal property registration and address management system. Field enumerators capture property information, location, GPS, and evidence photographs. Kebele Officers verify records, City Officers complete final approval, and the system generates an official address code.

Address code format: JIM-KB{KebeleCode}-ST{StreetCode}-BL{BlockCode}-HN{HouseNumber} .
Example: JIM-KB01-ABAJIF-BL01-HN101 .

2. Functional requirements

ID	Capability	Expected behavior	Primary users
FR-01	Authentication	Sign in, sign out, session validation, and protected application routes.	All users
FR-02	Password recovery	Request a reset by username, send a time-limited reset link by email, and allow a one-time password change.	All users
FR-03	Role-based access	Show and enforce functions according to Administrator, City Officer, Kebele Officer, Enumerator, and Viewer roles.	System
FR-04	Master data	Create, edit, view, and administer kebeles, streets, and blocks. Active location values are used during property capture.	Administrator, City Officer; limited view for other roles
FR-05	Property registration	Capture owner, contact, property type, ownership, building details, kebele, street, block, house number, and remarks.	Enumerator, Administrator
FR-06	GPS capture	Capture current latitude and longitude from a supported device/browser or enter coordinates where permitted.	Enumerator
FR-07	Evidence photos	Upload and classify front view, side view, business sign, document, or other photos. A front-view photo is required for submission.	Enumerator, Administrator
FR-08	Duplicate protection	Check likely duplicate property information before creating a new record.	Enumerator, reviewers
FR-09	Offline field work	Save field records locally as drafts while offline and sync them when connectivity returns.	Enumerator
FR-10	Approval workflow	Submit for verification, verify at kebele level, approve at city level, reject with a required reason, and resubmit corrected records.	Enumerator, Kebele Officer, City Officer
FR-11	Official addressing	Generate and display the official address code when City Officer approval is completed.	City Officer, Administrator; read-only visibility for others
FR-12	Notifications	Send two separate SMS alerts to the owner: after Kebele verification and after final City approval. Delivery requires valid Africa's Talking configuration.	Property owner; system operators

FR-13	Dashboard and analytics	Show totals, status distribution, kebele grouping, recent properties, trend information, and enumerator performance.	All authenticated users according to access
FR-14	GIS map	Display properties with GPS coordinates on an interactive Leaflet/OpenStreetMap map and link map locations to property records.	All authenticated users
FR-15	Reports	Filter property data, view address and property reports, and export report results where enabled.	All authenticated users according to access
FR-16	Audit trail	Record important property, approval, user, and report actions for accountability and review.	Administrator, City Officer; review access for operational roles
FR-17	User administration	Create, search, filter, update, activate/deactivate, and delete users. Kebele assignment is required for Kebele Officer and Enumerator roles.	Administrator; City Officer has management visibility

3. Non-functional requirements

Quality area	Requirement	Operational measure / acceptance guidance
Security	Protected routes, role checks, hashed passwords, bearer-token authentication, time-limited password reset tokens, and secret values kept outside source code.	Test every role against both permitted and denied actions; rotate production secrets before go-live.
Availability	Web application and API should run as separate restartable services with a health endpoint.	Monitor <code>/api/healthz</code> ; restart a failed service without changing stored data.
Performance	Lists, dashboard summaries, and filters should remain responsive for the expected municipal dataset.	Use pagination, indexed status/kebele/property fields, and report filters rather than loading all records unnecessarily.
Data integrity	Required owner and location fields, approval status transitions, rejection reasons, and database relationships must be validated.	Do not approve a record that is missing required evidence or review information.
Usability	Users should complete their common task with clear labels, visible status, and actionable error messages.	Enumerator can save offline; reviewer can understand the next action from the property status.
Mobile readiness	Field collection must work on a phone-sized screen and tolerate intermittent connectivity.	Test GPS permission, camera/photo upload, offline draft saving, and later synchronization on the field device.
Maintainability	API contract, generated client types, shared database schema, and frontend modules should remain synchronized.	Regenerate API clients after OpenAPI changes and run type checking before release.
Auditability	Approvals, rejection reasons, user actions, and exports should be traceable to an actor and timestamp.	Review the Audit Logs page during monthly data-quality checks.
Scalability	Support more kebeles, streets, blocks, users, photos, and properties without changing the business process.	Keep media storage and database backups independent from the application process.
Interoperability		

Use HTTP/JSON APIs, OpenAPI documentation, PostgreSQL, standard image uploads, OpenStreetMap tiles, and Africa's Talking SMS integration.

Document credentials and external service configuration per environment.

4. Acceptance checklist

Registration

- Owner and phone are captured.
- Kebele, street, block, and house number are valid.
- GPS and front photo are present before submission.
- Duplicate warning is reviewed.

Review

- Kebele Officer verifies or rejects with a reason.
- City Officer approves or rejects with a reason.
- Approved records have an address code.
- SMS events are logged operationally.

Operations

- Backups are scheduled.
- Health check is monitored.
- Demo passwords are changed.
- Production SMS and email credentials are tested.

Controlled reference document. Update this document when the approved workflow, roles, integrations, or deployment configuration changes.